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ORIGINAL ARTICLE

Live and prerecorded popular music consumption

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Abstract Changing consumption habits have rearranged the popular music market in the last decade, and a pattern in which live music attendance gets an increasing share of the market has emerged. This work analyzes the demand for the popular music sector considering its double dimension as supplier of live concerts and prerecorded music. We use the 2006/2007 wave of Spain's *Survey on Habits and Cultural Practices*, and estimate a bivariate probit model for attendance to live concerts and the purchase of prerecorded music. Results allow us to describe the profile of the average and frequent consumer in both markets, which shows some similarities—gender effects and the role of cultural capital—but also striking differences—time restrictions and relation to economic activity, and the use of technology. Finally, we find evidence of demand complementarities, with a direct causal link from prerecorded music to live attendance that helps explain recent institutional changes.

Keywords Live and prerecorded popular music · Participation · Audiences · Cultural demand · Cultural capital · Bivariate probit model

JEL Classification D12 · C25 · Z10 · Z11

> Although *Columbia* probably thinks otherwise, how the album fares commercially is almost incidental. With or without the CD business, Prince

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gets to keep making music in arenas, in clubs, in the studio. Fans buy concert tickets, companies rent his panache, pleasure is shared. It's a party that can go on a long time.*The Once and Future Prince*, by Jon Pareles. New York Times, July 29 2007

1 Introduction

New consumption patterns have shaken up the music industry. Faced with an ongoing process of structural change, cuts in prerecorded music sales have gone hand in hand with the increasing relevance of the live market as a source of revenues for artists and record labels.

In the past, live performances were a way of promoting sales of prerecorded music, a role no longer relevant as the diminishing share of the prerecorded music sector shows: the rise of illicit file-sharing, and the fact that legal digital downloads allow consumers to buy single tracks instead of albums explain the reduction of the profitability of this market. These demand-led changes have had a significant impact on the supply side of the music industry. Some salient features are: (1) the consolidation and vertical integration process taking place in the live sector, the *Live Nation-TicketMaster* merger being the most significant due to its global clout; (2) the expansion of long-term contracts, once the domain of prerecorded music, to live performances, such as the 12 year deal between U2 and *Live Nation*; or (3) the all-inclusive rights or "360 degree contracts" that record labels are offering to new acts or renegotiating with established ones. Overall this trend has led to more touring (The Economist 2010) and to a rise in ticket prices (Krueger 2005; Black et al. 2007).[1]

Some of these global trends are also observed at a local level in the Spanish popular music market. The recording industry, severely hit by the decline in demand, has adapted to the new environment by offering all-inclusive management services to artists in an attempt to capture part of the revenue of live performances.[2] With regard to the live sector, and after a period of remarkable growth in the demand for summer music festivals, competition has led to a reorganization and consolidation, with some festivals being driven out of the market and well-established ones aiming at expanding to new locations.[3] Last, trends in both markets

[1] Data by the trade organization of the recording industry (IFPI) shows that between 2000 and 2005 sales of prerecorded music CDs in OECD countries experienced an average annual growth rate of −3.57%, see http://www.ifpi.org/content/section_statistics. While figures for live music revenue are more fragmented, using Black et al. (2007) data for the US we find that between 1997 and 2005 total revenue in the live sector has experienced an average annual growth of 11.5%. Finally, the most recent IFPI figures reveal that global performance rights revenues have shown a steady growth since 2003.

[2] A trend that was advanced by independent labels such as *Subterfuge.com* that in the early 2000s started offering management services to their acts.

[3] Such as *Summercase*, a twin festival taking place in Barcelona and Madrid, entering the market in 2006 with an aggressive strategy (scheduled to coincide with the biggest summer festival) and that after its third edition is indefinitely "on hold"; or *Sonar* which usually takes place at Barcelona but in the 2010 edition took place also at Santiago de Compostela.

follow those found worldwide with a significant decline of the market share of prerecorded music.

Using available data for Spain, total revenue in the market for live concerts between 2000 and 2005 increased by an average annual rate of 11.1%. Over the same period, prerecorded music sales fell at an average annual rate of –4.4%.[4] In less than a decade, the live sector has doubled its overall market share from roughly a 14% in 2000 to a 28% in 2005. Even though more recent data are not available, this trend is likely to have been sustained as declining figures for the prerecorded market anticipate.

The above evidence sketches a picture of a changing market in which shifts in consumer demand for popular music (towards live performances) help explain observed aggregate behavior. Moreover, intuition suggests there is a link between both sectors with changes in one of them affecting the other. Our work aims at undertaking the analysis of the demand for (live and prerecorded) popular music in Spain. Specifically, we aim to identify different consumer profiles that will allow us to single out the determinants underlying market choices. In so doing, we reckon that both sectors cater to popular music consumers, with some of them participating in both the live and the prerecorded music market; hence, the need to account for potential interdependences. By using data from the 2006/2007 wave of the *Survey on Habits and Cultural Practices* (SHCP2006),[5] we jointly model the decision of attending a live performance and of purchasing prerecorded music as two binary variables, referred as attendance and purchase.

The main contribution of this work is the joint study of the demand for live and prerecorded music. As it has been noted, links between both markets may exist which calls for a combined consideration of both sides of the popular music market. We consider it as a novel contribution in that, to the best of our knowledge, no attempt has been made of an economic analysis of the music sector combining both dimensions and analyzing their interdependences.

Two main results emerge. First, a detailed description of the average and frequent consumer in the live and prerecorded music market. As recent evidence suggests, there are significant differences between consumers in both markets. Moreover, we also show that there are differences that determine the frequency of consumer participation in either market.

Second, the joint bivariate probit specification allows us to take into account the analysis of links and complementarities between both markets. These emerge as a result of two mechanisms: a direct one, based on the causality of one dependent variable on the other, and an indirect one, based on the correlation of the unobserved heterogeneity. Evidence supports a causal link between prerecorded music purchases and concert attendance, as well as a different correlation pattern between unobserved components for different consumer profiles.

[4] Data for live concerts from the main Spanish collecting society (SGAE, see its yearbook in http://www.artenetsgae.com/), and data for prerecorded music sales from the International Federation of the Phonographic Industry. Note that data on live revenue are available up to 2005.

[5] Elaborated by Ministerio de Cultura. Timing, sampling methodology and main results are available at http://www.mcu.es/estadisticas/MC/EHC/2006/Presentacion.html.

The structure of the work is the following. First, we analyze the literature on popular music consumption identifying three main avenues of research: on participation, on intellectual property rights (and its infringement) in the consumption of prerecorded music, and on live attendance and ticket price dynamics. Then we describe the conceptual framework, present the data set and a description of the variables. This is followed by a detailed presentation of the main results. We close with a discussion of the results including the analysis of the economic rationale of specific practices in the industry that are becoming current, such as new contractual relations.

2 Music consumer and music consumption

2.1 Literature review

From an economic standpoint observed market outcomes are the result of individual choices, incentives and constraints. Two preeminent and intertwined features of individual choice in the cultural domain are the low level of participation and the determinant role of previous cultural engagement or cultural capital (Throsby 1994; Seaman 2006). As Ateca-Amestoy (2008) puts it, a greater stock of cultural capital makes "individuals [...] more productive agents when they fulfill their cultural appreciation needs".

These traits are not absent from the discussion in the music sector, and participation has been analyzed in the music industry and classical music performance (Prieto-Rodriguez and Fernandez-Blanco 2000; Favaro and Frateschi 2007), with cultural capital, usually proxied by education and active engagement in the arts, being a key determinant in most studies. The analysis of participation leads to the definition of consumer profiles, which in the case of music consumption has been mainly undertaken from the sociological and the economic approach.

The sociological approach defines consumer behavior departing from status and social class variables and has traditionally considered cultural consumption as an individual's status indicator (DiMaggio, 1990; Peterson 1997). Linking social class to specific consumption patterns allows researchers to define a hierarchy of cultural manifestations, and the differentiation between "highbrow" and "popular" culture and their consumption. A differentiation that appears linked to an individual's relative position in the social ladder.

However, Peterson's studies in the 1990s—see for instance Peterson and Kern (1996)—show the appearance of an interesting pattern in which the link between the consumption of specific cultural manifestations and social class weakens. This leads to the emergence of a "high status" *omnivore consumer* that broadens its cultural engagement and, therefore, does not restrict her consumption to highbrow culture. Most recent literature on cultural consumption aims at testing the omnivore hypothesis and its determinants.

Along this line, Fisher and Preece (2003) study Canadian audiences and show that education and income have a significant and positive effect on "highbrow univores" and "omnivores", with gender being a feature of the former, and age

affecting negatively the latter. Meanwhile van Eijck (2001) offers an analysis of audiences in the Netherlands. His work clusters individuals in terms of their musical tastes and finds that consuming popular music is negatively related to age, education and active music participation, while gender or occupational status having no significant effect. On the other hand, omnivores are mainly men, young, educated, and active in music activities.

The economic approach to participation departs from the utility-maximizer behavior of individuals in markets. While music consumption can be analyzed at different levels of aggregation, data at the individual level (survey data) are likely to give a better picture of this maximization process. In this vein, some works explain classical and popular music consumption: Prieto-Rodriguez and Fernandez-Blanco (2000) consider the choice of listening to classical/popular music, and Favaro and Frateschi (2007) distinguish the equilibrium choice sets for different consumers, depending on whether they are univores (consumers of classical or popular music only), or omnivores (consumers of both types of music). Both papers are based in the Levy-Garboua and Montmarquette (1996) model for theater demand and, given the specificity of the data, introduce discrete choice models where individual market choices are reflected in a set of binary variables.

Prieto-Rodriguez and Fernandez-Blanco (2000) identify aspects that consumers of classical and popular music share, what they call an innate taste for music. In their work, the authors find a negative age effect and no evidence of a gender effect, while time restrictions are relevant when explaining consumption habits: time devoted to household tasks has a positive influence on popular music while family responsibilities reduce the likelihood of listening both classical and popular music. Finally, the authors find that education has a nonlinear effect on popular music consumption while a positive effect for classical music.

Favaro and Frateschi (2007) segment consumers in three groups: classical and popular music consumers, i.e. omnivorous; only popular music consumers; and only classical music consumers. Then they identify the factors that influence the likelihood of belonging to any of these groups. The authors find that age and gender (being female) is positively related to omnivorous preferences while negatively related to only popular music consumption; education has a positive effect for both groups; finally time availability is again a determinant for concert attendance in both groups, while there is mixed evidence with respect to occupational status.

In addition to participation, there are other strands in the literature related to music consumption. Specific to the prerecorded music market is the one on the copyright industries and the impact of copying technologies and the internet on consumption. The use of the internet allows consumers to access a wide array of contents, information and knowledge about music. As internet downloads directly compete with prerecorded music, the literature on intellectual property rights in the cultural industries has identified two main effects of file-sharing on prerecorded music consumption: substitution or crowding-out effects, and positive external effects usually under the label of exposition effects.

Measuring the effect of the internet on music consumption has been controversial and findings are ambiguous. Three examples are Zentner (2006), Montoro and Cuadrado (2008), and Oberholzer and Strumpf (2007); while the first two works find

a negative net effect of file sharing on prerecorded music consumption, the latter concludes that empirical evidence does not support any significant effect of internet downloads on music purchases. Interestingly, and to our best knowledge, the literature has been focused on the demand for prerecorded music disregarding potential cross effects on concert attendance.

To conclude, a third line of research that has generated some interesting insights from alternative standpoints, deals with the analysis of live popular music. Earl (2001) analyzes the behavioral economics of live popular music consumption. The author states that the demand for live music cannot be explained as the demand for an information good, as there are more efficient ways of gathering and of processing the same bits of information, for instance by watching a DVD of the same performance. As concert attendance involves costs that are not present when consuming prerecorded music,[6] attending live performances is a rather inefficient way of music consumption. In order to solve the apparent paradox, the author points out to several factors explaining demand, such as risk attitudes, curiosity and sampling, and the social and ritual dimension of live music consumption. In short, the author points out that the demand for live music should be considered as something different from the demand for music as a flow of aural and visual information.

In a practical sense, and from a pure economic standpoint, Earl's thesis raises two interesting questions, namely that of the nature of the relationship between recorded and live music consumption, whether they are substitutives or complements, and the underlying causes of attendance. If cultural needs are satisfied by either market and consumers face a choice between live and prerecorded music, then substitution prevails. However, if live music is more than just music as evidence seems to point out, there may be complementarities and links between both markets that may explain a positive correlation. On the other hand, there are differences between consumers of live concerts and of prerecorded music with each sector catering to different demand segments.

Also focused on the live sector but from a different perspective, other authors study the price dynamics of the US live market. Krueger (2005) takes ticket prices and finds that from 1996 to 2003 average concert price increased by 82%, compared to the CPI increase by 17%. After discarding other explanations (the potential crowding out of the secondary ticket market, rising superstar effects, Baumol and Bowen's disease, or increased concentration in the industry), the author tentatively concludes that the decline in complementarities between the prerecorded and the live market, what he coins as the *Bowie theory*,[7] explains the surge in concert prices.

[6] Earl (2001) identifies eight components in the total cost of attending a concert. Quoting the author, these are: "transport-related costs; child-related costs; poor sound quality and excessive volume; difficulties in seeing the performers; disadvantages of social consumption; undesired supporting artists; limited editing opportunities [i.e., inability of editing out less attractive parts of the show]; monopolistic suppliers of food and drink."

[7] After an interview in the *New York Times* in which, among other things, the artist reflects about the Internet and the changes it has brought to the music industry. At one point, he states "[...]You'd better be prepared for doing a lot of touring because that's really the only unique situation that's going to be left." See Pareles (2002).

Similarly Black et al. (2007) analyze the Top 100 concert tours in the US from 1997 to 2005. Findings are consistent with previous research in the sector, namely the escalation of price tickets and the stagnation of sales that yield an inelastic demand for the Top 100 tours.

Finally, and from a marketing perspective, Minor et al. (2004) identify the elements that drive customer satisfaction with live performances. The authors identify six factors; interestingly musical ability and musician appearance, two specific features of live music, ranked on top. However, one should note that, given the nature of the study (focused in small venues), these factors apply for performers other than superstars.

The research in this paper shares some common elements with Prieto-Rodriguez and Fernandez-Blanco (2000), both in its main objective, identifying the profile of the popular music consumer, and methodology, a bivariate probit model. Specific to our work, and the key contribution of this paper, is the joint consideration of the prerecorded and live popular music sector. In other words, we analyze participation as the joint consumption of popular live and prerecorded music consumption.

In addition, we carry out some specific tests linked with the previously reviewed literature. Fist, we test the effect of the increased accessibility to prerecorded music both in prerecorded and live music consumption, i.e. we test for substitution and exposition effects in both markets. Second, by using a set of dummy variables that reveal restrictions individuals face when consuming live music, we put forward and test two of the hypotheses in Earl's paper, namely the social-gathering nature of live music consumption and the potential substitutability of live and prerecorded music. Finally, we aim at testing Krueger's conjecture, the *Bowie hypothesis*, by analyzing the direction of the causal link between the live and prerecorded music market.

2.2 Conceptual framework

We use a random utility model as the conceptual building block for individual decision making in the market for prerecorded and live popular music. From a standard economic perspective and assuming that observed outcomes reveal alternatives that maximize utility, it allows us to link observed choices to the maximization of expected utility.

Let us consider that the utility derived from popular music consumption arises from attending live concerts (u_1) and/or purchasing prerecorded music (u_2). We assume these are linear in the characteristics of the choice alternatives and of the individuals. These are collected in a vector of explanatory variables x'_i. Then:

$$u_1 = x_1'\beta + \epsilon_1 \tag{1}$$

and

$$u_2 = x_2'\beta + \epsilon_2 \tag{2}$$

where ε_1 and ε_2 follow a bivariate normal distribution with

$$E[\epsilon_1|x_1, x_2] = E[\epsilon_2|x_1, x_2] = 0, \quad \mathrm{Var}[\epsilon_1|x_1, x_2] = \mathrm{Var}[\epsilon_2|x_1, x_2] = 1,$$

and

$$\mathrm{Cov}[\epsilon_1, \epsilon_2 | x_1, x_2] = \rho.$$

Let the utility of the alternatives to popular music consumption be $u^0{}_i$, with $i = 1,2$. Whenever an individual is observed to participate in activity i, attendance to live concerts or purchase of prerecorded music, it must be that $u_i > u_i^0$.[8]

In this setup, we can model observed choices using two binary variables: y_1 and y_2. These measure whether an individual attends to a live concert or purchases prerecorded music, respectively. Attendance or purchases are observed if:

$$y_1 = 1 \quad \text{if} \quad u_1 > u_1^0 \tag{3}$$

$$y_2 = 1 \quad \text{if} \quad u_2 > u_2^0. \tag{4}$$

In short, expressions (1)–(4) can be estimated as a bivariate probit specification, which allow for the correlation between y_1 and y_2. This emerges from:

1. The dependence of both variables on correlated observed variables.
2. The correlation of the unobserved component, i.e. the correlation between ε_1 and ε_2, measured by ρ.

Besides, we can consider an additional source of correlation: a direct causal relationship between y_1 and y_2. The causal link implies that one market has a direct influence on the other. In order to test for this possibility, we include the following specification:

$$u_i = \beta' x_i + \alpha_j y_j + \epsilon_i. \tag{5}$$

with $i \neq j$. For instance, this setup allows for a statistically significant direct effect of live attendance to purchases (purchases to live attendance) if $\alpha_1 \neq 0$ ($\alpha_2 \neq 0$).

3 Analysis and description of the data

3.1 The survey

The empirical analysis relies on the information provided by the SHCP2006. It is a survey-based research conducted by the Spanish Ministry of Culture in partnership with the National Statistics Agency. It collects data from a sample of roughly 16,000 individuals (representative of the Spanish population over 15), about their participation in the arts and cultural activities over the previous 12 months.[9]

The main aim of the survey is twofold. First, to evaluate the evolution of the main indicators of cultural habits and practices in Spain and the frequency of participation. Second, to have a deeper understanding of some specific patterns of

[8] As noted in a referee report, it is debatable whether actual choices are based on a maximization process. Nevertheless, rational choice is not a necessary condition and alternative decision rules may fit this setup. For instance, we can model consumer choice based on satisfying choices by defining u_i^0 as a minimum utility level to enter one of the markets.

[9] The survey uses a two-stage sampling method with stratified primary sampling units. We use sample weights in the empirical work.

cultural consumption, specially for information goods and services protected under intellectual property.[10]

The questionnaire is structured in sections that collect general information and data about participation and consumption in specific cultural manifestations. We have processed information from most of these sections. Dependent variables are built from the revealed information about participation in live concerts and the purchase of prerecorded music. More specifically, these are defined as the binary answers to "[the respondent] attended to a live popular music concert at least once over the last year/3 months" and "[the respondent] purchased prerecorded music at least once over the last year/3 months". The time frame of reference (1 year or 3 months) allows us to capture differences between frequent consumers, those that bought or attended at least once over last 3 months, and average consumers, those that attend or buy at least once a year. We use the label average as the last group includes the former, and indeed what we are doing is to average consumption over frequent and less-frequent consumers.[11]

Two comments apply with regard to purchases of prerecorded music. First, and as our primary interest was on popular music, we have adjusted the purchase variable to exclude consumers buying classical music and opera. Second, the survey considers prerecorded music in a broad sense. It is not restricted to the physical format, or compact discs, but also includes downloads from online stores and other digital platforms such as iTunes.

Table 1 shows a cross tabulation of average consumers in both markets. From its inspection, we find a low level of participation, a rather consistent result with the literature in cultural attendance: roughly 25% of the sample attended to a concert or purchased music over last year, while less than 12% engaged in both activities. Cross cultural comparisons stress this regularity: data for Canada from Fisher and Preece (2003) and Italy from Favaro and Frateschi (2007) show, respectively, a 72.5 and 75.9% share of individuals not participating at all. This is a remarkable finding, once we note that we are dealing with manifestations of popular culture.

Table 2 shows the same cross tabulation for frequent consumers in both markets. While there is a slight decrease of CD buyers, now 18.2% of the sample, the share of frequent concert-goers drops to roughly 12% of the sample. More significant is the fact that only a 4.3% of the sample engages in both activities, which could suggest different determinants affecting frequent and average consumption. However one must note that not all surveys were taken in the same term of the year. If live performances show a seasonal pattern, then the likelihood of attending a concert in the past 3 months will be affected by the month in which the survey is answered. This makes it necessary to account for this term effect when making comparisons between both consumer types.

[10] From a quantitative standpoint, the survey design and data collection procedure offers the researcher a view of cultural consumption for the population over 15 in Spain. However, the good quality and breadth of the data comes at a cost. As one referee noted, designing the survey questionnaire would have allowed us for more specific questions about music consumption. Nevertheless to obtain a representative sample of the Spanish cultural consumer in this case would have been prohibitive.

[11] It would not have been accurate to estimate an expression for infrequent consumers. Note that in our setup, infrequent consumption is the opposite of both frequent and no consumption at all. Hence, the meaning of the zeroes in expressions (3)–(4) would not be univocal, leading to inconsistent estimates.

Table 1 Attendance to live music concerts and purchase of prerecorded music: at least once last year

		Purchase		Total
		No	Yes	
Attendance	No	8,884	1,983	10,867
	Yes	2,203	1,752	3,955
	Total	11,087	3,735	14,822

Crosstabulation of EHPC2006 data

Table 2 Attendance to live music concerts and purchase of prerecorded music: at least once last 3 months

		Purchase		Total
		No	Yes	
Attendance	No	10,978	2,052	13,030
	Yes	1,149	643	1,792
	Total	12,127	2,695	14,822

Crosstabulation of EHPC2006 data

To estimate expressions (1)–(2), we choose a set of explanatory variables, all of them related to the reviewed literature. As has been highlighted, socio-demographic and economic factors, education, active participation and involvement (cultural capital), and substitutes and complements potentially affect cultural consumption. We group these explanatory variables in five blocks:

1. Socioeconomic and geographical variables.
2. Cultural capital and related variables.
3. Physical capital and cultural equipment.
4. Use of the internet and file-sharing networks.
5. Revealed restrictions to attendance and purchase.

Table 3 lists all the variables, gives a brief description, when needed, and shows basic descriptive statistics and the expected effect on the endogenous variables (attendance/purchases).

It is worth noting that the data set does not include income. If, as one should expect, its effect on consumption is significant, some of the variables in our list will capture it making their interpretation tricky. Candidates are those variables related to education level, physical equipment of the household and labor market situation of the survey-taker. Besides, a set of regional variables is included to proxy for spatial income differences.

3.2 Explanatory variables

Next we discuss in detail each of the previous blocks of covariates, and their expected impact on participation.

Table 3 Variables description and main statistics

Variable	Type	Description	Mean	SD	Effect
Endogenous variables—average consumer					
Live	D	Attended live concerts (last year)	0.2668	0.4423	
CD	D	Purchased music (last year)	0.2519	0.4342	
Endogenous variables—frequent consumer					
Live	D	Attended live (last 3 months)	0.1209	0.3260	
CD	D	Purchased music (last 3 months)	0.1818	0.3857	
Personal characteristics					
Female	D		0.5217	0.4995	?/?
Age	C		47.4021	19.1512	−/ −
Single	D		0.3556	0.4787	+/?
Child15	N	Number of children in household below 15	0.3299	0.7063	−/?
Child18	N	Number of children in household between 15 and 18	0.0855	0.3339	−/?
Labor market and economic activity					
Works	D		0.4988	0.5000	?/?
Unemployed	D	Unemployed: had previous job	0.0549	0.2278	?/?
Looking	D	Unemployed: looking for first job	0.0068	0.0819	?/?
Student	D		0.0900	0.2862	+/?
Retired	D		0.1785	0.3830	?/?
Domestic	D	Househusband/housewife	0.1556	0.3625	?/?
Education					
Vocational	D	Vocational education	0.1167	0.3210	?/?
High school	D	Intermediate studies	0.1263	0.3322	?/?
University	D	Higher education	0.1533	0.3603	?/?
Active practice and participation					
Active	D	Active engagement in the practice of a cultural activity	0.3688	0.4825	+/+
Courses	D	Took course(s) in an art/cultural topic	0.0108	0.1033	+/+
Heritage	D	Heritage participation	0.4678	0.4990	+/+
Performing	D	Performing arts participation	0.2559	0.4364	+/+
Cinema	D	Watched films on theater	0.5086	0.4999	+/+
Video	D	Watched films on Video/DVD	0.312	0.4633	+/+
ReadingHrs	C	Weekly hours reading	0.9859	1.7194	+/+
RadioHrs	C	Weekly hours listening to radio	4.6281	5.4291	+/+
TVHrs	C	Weekly hours watching TV	8.1003	5.588	+/+
Restrictions: why don't you attend/buy more?					
PriceRestL	D	High price of live concerts	0.2204	0.4145	+/
QuantityRestL	D	Difficult to get tickets	0.0215	0.1451	+/
VarietyRestL	D	Not many opportunities (little variety)	0.2003	0.4002	+/
DVDTVRestL	D	Prefer watch on TV	0.0337	0.1806	−/
SocialRestL	D	Cannot find a companion for	0.0082	0.0904	−/
PriceRestCD	D	High price of CDs	0.4981	0.5000	/+

Table 3 continued

Variable	Type	Description	Mean	SD	Effect
Physical capital					
RadioEq	D	Has radio equipment	0.9736	0.1603	+/+
CDEq	D	Has CD player	0.8079	0.3940	+/+
DiskmanEq	D	Has portable CD player	0.4648	0.4988	+/+
TapeEq	D	Has tape player	0.6994	0.4585	+/+
VynilEq	D	Has turntable	0.3091	0.4622	+/+
MP3Eq	D	Has MP3 player	0.3890	0.4875	+/+
Instruments	N	Number of music instruments	0.4572	1.3435	+/+
VideoGameEq	D	Has video game console	0.2313	0.4217	+/+
VideoCamEq	D	Has camcorder	0.3027	0.4594	+/+
TVEq	N	Number of TV sets in household	2.1064	1.0909	+/+
PhotoEq	N	Number of photograph cameras in household	1.1397	1.5437	+/+
Internet					
InternetHrs	C	Weekly hours using the Internet	1.7022	5.1262	?/?
P2P	D	User of file-sharing networks	0.1067	0.3087	?/?

SD standard deviation; *D* binary variable; *N* count variable; *C* continuous variable

+/?: Variable has a positive effect on live attendance and an ambiguous effect on CD puchases

3.2.1 Socioeconomic and geographic variables

Personal characteristics. Personal characteristics include traits such as gender, age, being single and children in the household, i.e. number of children in the household below 15 and between 15 and 18. We do not have any a priori hypothesis on the specific impact of gender on popular music consumption, and previous works on the popular music sector are ambiguous. In the literature review, only Favaro and Frateschi (2007) find a significant and negative gender effect on the consumption of popular music.

Pop and rock music are mainly young-oriented cultural manifestations.[12] Correspondingly we expect age to affect negatively both endogenous variables. However, and in order to test for nonlinearities, we include age and its square. Being single could have a positive effect on attendance by reducing decision-making costs and relaxing the time constraint, while no effect is hypothesized on purchases. Having children to care for is a handicap for attendance, at least for those below 15, as it tightens up the time restriction an individual faces. Nevertheless, its effect is not clear for purchasing music.

[12] At least one can say that popular music consumers are younger in age, as the raw comparison of the mean age for the average participant and non-participant in both markets shows: roughly 34 years compared to 51 in the live market, and 35–52 in the prerecorded one. Interestingly, the 2002 wave of the survey shows that the mean age was 30 compared to 47.4 for live attendance, and 33 compared to 48.6 for prerecorded purchases. As one referee notes, one ought to recognize that the success of nostalgia-based tours by established acts responds to the aging of the consumer of popular music.

Labor market and economic activity. The status of an individual in the labor market defines her economic well-being and opportunity cost of time. Therefore, most variables in this section will capture an individual's time and income availability.

Assuming that popular music as a leisure activity is primarily targeted to young people, then its consumption should be correlated with being a student. On the other hand, if leisure time availability is one of the main restrictions for cultural participation, full-time students will have a lower time restriction for both activities. Obviously, these positive age and time effects can be partially offset if there is an income effect, as we expect income to be negatively correlated with being a student. Overall we expect the net effect to be positive for attendance.

Working, either as an employer or employee, relaxes the budget constraint of an individual, increasing the probability of attendance and purchase. However, the time restriction for attending a concert or listening to music plays in the opposite direction, and therefore, the expected effect is ambiguous. Being unemployed reduces the choice set of an individual but may shift the demand for time-intensive activities, leading to a negative income effect and a positive time effect. Depending on the relevance attached to each item, income vs. time, cultural consumption might be more or less likely. Note that we distinguish between unemployed with and without a previous job. One expects the latter demand segment, i.e. unemployed looking for the first job, to be mainly composed of young individuals with plenty of available time.

In a similar vein, we expect those out of the labor market, retired and househusbands or housewives, to have less disposable income but more time availability. Overall, the net effect is undetermined.

Geographical variables. All estimations include—although results are not shown—one dummy for each of the 19 Spanish regions, and a set of variables that reflect the urban environment. Our aim is to capture geographical demand and supply factors.

With regard to urban environment, the survey does not offer a clear distinction based on a population size criterion. For communities are classified by size—more than 100,000 inhabitants, between 50,000 and 100,000, between 10,000 and 50,000, and less than 10,000 inhabitants, except province capitals that are a category on their own. Given the heterogeneity among this last category—combining big cities such as Madrid (over 3 million inhabitants) or Barcelona (around 1,600,000), with significantly smaller ones as Teruel (35,000 inhabitants), any attempt to capture community size with this variable is beyond scope. This is the main reason why, although included in the estimation, results are not shown.

By and large, different regional and local economic conditions in our estimations, which we expect to be be correlated with income and supply factors, are in this way incorporated in our estimations.

3.2.2 Cultural capital and cultural participation

This block includes education level, the participation in other cultural manifestations, and active cultural practice.

Education. Most authors link cultural capital accumulation with the education level as it helps explaining the process of acquisition of cultural tastes. The underlying rationale is that the higher the education level the higher the probability of cultural consumption, as more educated individuals acquire the competences for the appreciation of culture and face a smaller shadow price when consuming art.

However, popular music is, by definition, popular, and generally little background knowledge is needed to enjoy this specific cultural manifestation. In fact we expect the frequency of consumption, but not participation, to be linked to the accumulation of knowledge and it need not be necessarily linked to education achievement. Hence, we assume no prior sign in the relationship of the maximum education level attained and the likelihood of consumption.

Active practice and participation. Variables in this set include active cultural practice—painting, writing, photography, video art, acting, dancing, singing, playing etc.—and participation in other cultural manifestations. Overall we expect the sign of these effects to be positive.

We also include weekly hours devoted to leisure reading, listening to radio and watching TV. Reading as a cultural practice may have a similar effect as previous variables. However, listening to radio and watching TV may have ambiguous effects. While both may increase knowledge about music in general and of live concerts and new releases in particular as promotion in the music sector resorts to TV and radio as the main channels, they can be also "lowbrow" leisure activities competing with popular music consumption. If we assume the role of the information diffusion being predominant, the expected net effect will be positive.

3.2.3 Physical capital

We have chosen several variables that are linked to the cultural equipment of the household. The expected net effect in all cases is positive, as these variables convey information about equipment but also about household income. Therefore, we should be cautious when interpreting these results.

3.2.4 Internet use

We include two variables that capture an individual's use of the internet: weekly hours using the internet (out of the workplace/study place); and being a user of file-sharing networks. The former tries to capture the use of the internet as a leisure-time activity, thus in competition with other leisure activities, and as an alternative way of music consumption, with sites such as YouTube offering popular music contents.[13] The latter aims at capturing new consumption patterns in prerecorded music.

In this last respect, music availability through the internet can crowd out prerecorded music consumption, or it may enhance it if external effects play a role

[13] As one referee noted the rise of YouTube may affect the demand for music, both prerecorded and live, in the same way as file sharing does. The potential emergence of substitution and exposition effects leads to an ambiguous net effect on music demand.

as consumers' demand is expanded by the discovery of new acts, music genres, etc. In brief, we check for crowding-out and exposition effects.

3.2.5 Restrictions on attendance

Here, we include the main reasons why individuals do not attend or purchase more due to price, quantity as well as variety restrictions. Note that these restrictions convey information about cultural practices and that by declaring that one would consume more if some specific restriction was relaxed could make one more likely, and not less, to participate. In short, better informed consumers will be more likely to give an affirmative response to this questions.

Moreover, we will test the substitution nature of live and prerecorded music consumption, and the social motivation for attendance. Both are found in Earl (2001) and allow us to identify individuals that see music merely as information, and individuals that find social gathering as one relevant reason for attendance. Indeed, as it will be seen, what we will be testing is if social motivations are strong enough to prevent attendance.

We do so by including two specific constraints. First, the answer to "I do not attend to live performances more frequently as I prefer to watch them on TV or DVD or listen to them in Radio". This allows us to identify to what extent live attendance and alternative, and less costly, ways of music consumption are substitutes. Second, social motivation to attendance is captured as the answer to " I do not attend to live performances more frequently as I do not have anyone to go with."

4 Results

Tables 4 and 5 show estimation results for the average and frequent consumers, respectively. Each presents three specifications, the main difference being that model 2 and 3 are recursive bivariate probit specifications, i.e. they include a (previously and jointly determined) endogenous variable in the right-hand side of one of the equations. In other words, models 2 and 3 in both tables correspond to the specification given by expression (5).

As usual, maximum likelihood estimation (MLE) is used. Note that even though the recursive specification includes an explanatory endogenous variable, minimum restrictions that guarantee the model is identified and MLE yields consistent estimates. As Greene (2007) notes, the endogeneity of one of the explanatory variables can be ignored in formulating the log-likelihood function.

Tables include estimated coefficients and their corresponding standard errors in parenthesis. We also include the estimated correlation coefficient, ρ, and a joint significance test, χ^2. Results show different profiles for consumers in both cultural manifestations. We start spelling out the average consumer and then single out the main differences between average and frequent consumers.

Table 4 Estimation results: average consumer

	Model 1		Model 2		Model 3	
Variables	Live	CD	Live	CD	Live	CD
Live				−0.1097 (0.1343)		
CD					0.8399** (0.1293)	
Female	−0.1224** (0.0325)	−0.1447** (0.0315)	−0.1225** (0.0324)	−0.1478** (0.0316)	−0.0835** (0.0328)	−0.1426** (0.0315)
Age	−0.0101 (0.0070)	0.0041 (0.0076)	−0.0104 (0.0070)	0.0031 (0.0077)	−0.0005 (0.0070)	0.0044 (0.0076)
Age^2	−0.0001** (0.0001)	−0.0003** (0.0001)	−0.0001* (0.0001)	−0.0003** (0.0001)	−0.0002** (0.0001)	−0.0003** (0.0001)
Works	0.2410* (0.1363)	0.3710** (0.1419)	0.2371* (0.1361)	0.3765** (0.1406)	0.1436 (0.1407)	0.3874** (0.1437)
Unemployed	0.1757 (0.1461)	0.2287 (0.1511)	0.1720 (0.1458)	0.2340 (0.1499)	0.1216 (0.1498)	0.2421 (0.1528)
Looking	0.2217 (0.2095)	0.1965 (0.2068)	0.2154 (0.2099)	0.2042 (0.2057)	0.1917 (0.2108)	0.2147 (0.2071)
Student	0.2305 (0.1468)	0.0854 (0.1515)	0.2249 (0.1469)	0.0941 (0.1506)	0.2400 (0.1499)	0.1002 (0.1532)
Retired	−0.0260 (0.1517)	0.3782** (0.1598)	−0.0320 (0.1517)	0.3769** (0.1585)	−0.1195 (0.1535)	0.3925** (0.1615)
Domestic	0.1698 (0.1445)	0.3489** (0.1491)	0.1648 (0.1444)	0.3530** (0.1477)	0.0713 (0.1481)	0.3695** (0.1508)
Single	0.2284** (0.0411)	0.0084 (0.0414)	0.2290** (0.0411)	0.0186 (0.0430)	0.2199** (0.0405)	0.0130 (0.0412)
Child15	−0.0905** (0.0242)	0.0240 (0.0216)	−0.0906** (0.0242)	0.0201 (0.0222)	−0.0999** (0.0238)	0.0253 (0.0217)
Child18	−0.0304 (0.0460)	0.0111 (0.0427)	−0.0296 (0.0459)	0.0099 (0.0426)	−0.0308 (0.0454)	0.0149 (0.0426)
High school	0.0695 (0.0450)	0.0685 (0.0441)	0.0708 (0.0451)	0.0706 (0.0440)	0.0432 (0.0452)	0.0658 (0.0441)
Vocational	0.1145** (0.0457)	0.0250 (0.0448)	0.1153** (0.0457)	0.0292 (0.0448)	0.0963** (0.0457)	0.0240 (0.0448)
University	0.0968** (0.0449)	−0.0123 (0.0454)	0.0974** (0.0449)	−0.0083 (0.0454)	0.0933** (0.0448)	−0.0117 (0.0453)
Active	0.2149** (0.0321)	0.1785** (0.0312)	0.2157** (0.0321)	0.1849** (0.0319)	0.1628** (0.0332)	0.1758** (0.0313)
Courses	0.2147 (0.1307)	0.0657 (0.1258)	0.2163* (0.1309)	0.0735 (0.1266)	0.1891 (0.1248)	0.0726 (0.1222)
Heritage	0.2949** (0.0351)	0.1910** (0.0338)	0.2952** (0.0351)	0.2004** (0.0355)	0.2414** (0.0364)	0.1870** (0.0340)
Performing	0.3861** (0.0361)	0.0717** (0.0356)	0.3863** (0.0361)	0.0858** (0.0396)	0.3612** (0.0366)	0.0651* (0.0358)
Cinema	0.1539** (0.0358)	0.2239** (0.0347)	0.1542** (0.0357)	0.2275** (0.0349)	0.0923** (0.0367)	0.2246** (0.0346)

Table 4 continued

	Model 1		Model 2		Model 3	
Variables	Live	CD	Live	CD	Live	CD
Video	0.0849** (0.0324)	0.1898** (0.0312)	0.0855** (0.0324)	0.1915** (0.0311)	0.0237 (0.0336)	0.1900** (0.0312)
ReadingHrs	−0.0097 (0.0086)	0.0246** (0.0084)	−0.0097 (0.0086)	0.0241** (0.0084)	−0.0153* (0.0086)	0.0251** (0.0083)
RadioHrs	0.0066** (0.0028)	0.0044 (0.0027)	0.0066** (0.0028)	0.0046* (0.0027)	0.0053* (0.0027)	0.0047* (0.0027)
TVHrs	0.0016 (0.0031)	0.0080** (0.0030)	0.0016 (0.0031)	0.0080** (0.0030)	−0.0008 (0.0031)	0.0076** (0.0030)
RadioEq	−0.0646 (0.1215)	0.3446** (0.1287)	−0.0644 (0.1213)	0.3395** (0.1292)	−0.1269 (0.1110)	0.3151** (0.1299)
CDEq	0.0796 (0.0633)	0.3811** (0.0684)	0.0807 (0.0634)	0.3818** (0.0682)	0.0444 (0.0623)	0.3869** (0.0680)
TapeEq	0.0271 (0.0414)	0.1548** (0.0406)	0.0266 (0.0414)	0.1543** (0.0405)	−0.0119 (0.0410)	0.1522** (0.0406)
VynilEq	0.0573* (0.0346)	−0.0306 (0.0335)	0.0579* (0.0346)	−0.0275 (0.0335)	0.0667* (0.0343)	−0.0257 (0.0336)
DiskmanEq	0.0014 (0.0362)	0.0595* (0.0351)	0.0015 (0.0361)	0.0594* (0.0351)	−0.0192 (0.0357)	0.0613* (0.0350)
MP3Eq	0.1029** (0.0359)	−0.0181 (0.0351)	0.1028** (0.0359)	−0.0142 (0.0353)	0.1024** (0.0356)	−0.0182 (0.0350)
VideoGameEq	0.1627** (0.0362)	0.0304 (0.0355)	0.1627** (0.0362)	0.0360 (0.0360)	0.1499** (0.0361)	0.0341 (0.0354)
Instruments	−0.0005 (0.0103)	0.0042 (0.0090)	−0.0005 (0.0103)	0.0043 (0.0091)	−0.0015 (0.0097)	0.0048 (0.0089)
VideoCamEq	−0.0222 (0.0337)	0.0089 (0.0325)	−0.0224 (0.0337)	0.0073 (0.0324)	−0.0230 (0.0334)	0.0086 (0.0325)
TVEq	0.0151 (0.0142)	0.0134 (0.0137)	0.0151 (0.0142)	0.0139 (0.0137)	0.0136 (0.0141)	0.0133 (0.0136)
PhotoEq	0.0110 (0.0091)	0.0156 (0.0097)	0.0110 (0.0093)	0.0159 (0.0098)	0.0075 (0.0086)	0.0151 (0.0096)
PriceRestL	0.3828** (0.0369)		0.3815** (0.0368)		0.3684** (0.0370)	
QuantityRestL	0.4474** (0.0901)		0.4428** (0.0898)		0.4417** (0.0889)	
VarietyRestL	0.5882** (0.0388)		0.5837** (0.0397)		0.5742** (0.0391)	
DVDTVRestL	−0.0436 (0.1043)		−0.0438 (0.1040)		−0.0404 (0.1017)	
SocialRestL	0.0366 (0.1836)		0.0385 (0.1828)		0.0395 (0.1807)	
PriceRestCD		0.2761** (0.0308)		0.2755** (0.0306)		0.3004** (0.0306)

Table 4 continued

Variables	Model 1		Model 2		Model 3	
	Live	CD	Live	CD	Live	CD
InternetHrs	−0.0012 (0.0027)	−0.0082** (0.0028)	−0.0013 (0.0027)	−0.0082** (0.0027)	0.0012 (0.0029)	−0.0086** (0.0028)
P2P	0.0382 (0.0484)	−0.1056** (0.0467)	0.0379 (0.0484)	−0.1029** (0.0468)	0.0635 (0.0477)	−0.1077** (0.0469)
ρ	0.1742**		0.2376**		−0.3286**	
χ^2	4,142.5122		4,082.6888		5,090.5620	
p-value	0.0000		0.0000		0.0000	

Standard errors in parenthesis. $^{**}P < 0.05$, $^{*}P < 0.1$

Table 5 Estimation results: frequent consumer

	Model 1		Model 2		Model 3	
Variables	Live	CD	Live	CD	Live	CD
Live				0.0030 (0.1736)		
CD					0.1741** (0.0645)	
Term 2	0.5931** (0.0488)	−0.0794* (0.0431)	0.5932** (0.0488)	−0.0798 (0.0495)	0.6004** (0.0490)	−0.1212** (0.0403)
Term 3	0.1168** (0.0527)	−0.0745* (0.0432)	0.1169** (0.0528)	−0.0745* (0.0433)	0.1214** (0.0528)	−0.1165** (0.0405)
Term 4	−0.1632** (0.0554)	−0.0193 (0.0430)	−0.1632** (0.0554)	−0.0192 (0.0435)	−0.1628** (0.0557)	−0.0813** (0.0406)
Female	−0.1605** (0.0381)	−0.1303** (0.0334)	−0.1605** (0.0382)	−0.1303** (0.0337)	−0.1550** (0.0383)	−0.1445** (0.0315)
Age	−0.0045 (0.0086)	0.0040 (0.0083)	−0.0045 (0.0086)	0.0040 (0.0083)	−0.0035 (0.0086)	0.0043 (0.0076)
Age^2	−0.0001 (0.0001)	−0.0003** (0.0001)	−0.0001 (0.0001)	−0.0003** (0.0001)	−0.0001 (0.0001)	−0.0003** (0.0001)
Works	0.5795** (0.2103)	0.5725** (0.1601)	0.5796** (0.2103)	0.5722** (0.1606)	0.5481** (0.2105)	0.3785** (0.1431)
Unemployed	0.6177** (0.2184)	0.4634** (0.1693)	0.6178** (0.2184)	0.4631** (0.1700)	0.5935** (0.2186)	0.2309 (0.1524)
Looking	0.7767** (0.2776)	0.2661 (0.2254)	0.7769** (0.2779)	0.2656 (0.2264)	0.7635** (0.2775)	0.1986 (0.2068)
Student	0.7055** (0.2177)	0.4270** (0.1694)	0.7056** (0.2179)	0.4266** (0.1711)	0.6850** (0.2179)	0.0896 (0.1526)
Retired	0.2893 (0.2337)	0.5612** (0.1792)	0.2893 (0.2337)	0.5610** (0.1793)	0.2538 (0.2336)	0.3916** (0.1611)
Domestic	0.5252** (0.2222)	0.5961** (0.1675)	0.5253** (0.2222)	0.5959** (0.1679)	0.4911** (0.2224)	0.3594** (0.1503)
Single	0.1658** (0.0502)	−0.0244 (0.0450)	0.1657** (0.0503)	−0.0245 (0.0459)	0.1692** (0.0502)	0.0122 (0.0414)
Child15	−0.1137** (0.0304)	0.0304 (0.0229)	−0.1137** (0.0304)	0.0305 (0.0233)	−0.1167** (0.0306)	0.0256 (0.0217)
Child18	0.0509 (0.0591)	0.0221 (0.0456)	0.0509 (0.0592)	0.0221 (0.0457)	0.0506 (0.0595)	0.0155 (0.0426)
High school	0.0521 (0.0516)	−0.0142 (0.0476)	0.0520 (0.0519)	−0.0143 (0.0477)	0.0515 (0.0518)	0.0696 (0.0442)
Vocational	0.1228** (0.0534)	0.0040 (0.0472)	0.1228** (0.0534)	0.0039 (0.0473)	0.1220** (0.0536)	0.0231 (0.0449)
University	0.0627 (0.0528)	−0.0102 (0.0487)	0.0627 (0.0529)	−0.0103 (0.0487)	0.0626 (0.0530)	−0.0119 (0.0454)
Active	0.2217** (0.0381)	0.1560** (0.0332)	0.2217** (0.0381)	0.1558** (0.0340)	0.2162** (0.0382)	0.1760** (0.0313)
Courses	0.1852 (0.1407)	0.1799 (0.1251)	0.1852 (0.1407)	0.1798 (0.1253)	0.1743 (0.1405)	0.0626 (0.1246)

Table 5 continued

Variables	Model 1		Model 2		Model 3	
	Live	CD	Live	CD	Live	CD
Heritage	0.2092** (0.0423)	0.1731** (0.0359)	0.2092** (0.0423)	0.1730** (0.0366)	0.2009** (0.0425)	0.1911** (0.0339)
Performing	0.2907** (0.0413)	0.0770** (0.0375)	0.2908** (0.0414)	0.0768* (0.0396)	0.2887** (0.0415)	0.0678* (0.0357)
Cinema	0.0958** (0.0440)	0.1657** (0.0374)	0.0958** (0.0440)	0.1657** (0.0375)	0.0879** (0.0442)	0.2199** (0.0347)
Video	0.0836** (0.0383)	0.2033** (0.0330)	0.0836** (0.0385)	0.2033** (0.0330)	0.0727* (0.0388)	0.1875** (0.0312)
ReadingHrs	0.0114 (0.0096)	0.0386** (0.0087)	0.0114 (0.0096)	0.0386** (0.0087)	0.0095 (0.0096)	0.0243** (0.0084)
RadioHrs	0.0103** (0.0034)	0.0090** (0.0028)	0.0103** (0.0034)	0.0090** (0.0029)	0.0098** (0.0034)	0.0043 (0.0027)
TVHrs	−0.0071* (0.0041)	0.0008 (0.0033)	−0.0071* (0.0041)	0.0008 (0.0033)	−0.0072* (0.0041)	0.0081** (0.0030)
RadioEq	0.1092 (0.1445)	0.1894 (0.1403)	0.1092 (0.1446)	0.1893 (0.1404)	0.1023 (0.1444)	0.3368** (0.1304)
CDEq	0.0804 (0.0863)	0.4227** (0.0794)	0.0803 (0.0865)	0.4227** (0.0794)	0.0722 (0.0868)	0.3901** (0.0686)
TapeEq	0.0596 (0.0517)	0.1610** (0.0438)	0.0596 (0.0517)	0.1610** (0.0438)	0.0518 (0.0518)	0.1547** (0.0407)
VynilEq	0.0771* (0.0407)	0.0050 (0.0352)	0.0771* (0.0409)	0.0049 (0.0354)	0.0769* (0.0409)	−0.0281 (0.0335)
DiskmanEq	0.0478 (0.0428)	0.0343 (0.0373)	0.0478 (0.0428)	0.0343 (0.0374)	0.0453 (0.0430)	0.0557 (0.0351)
MP3Eq	0.0200 (0.0427)	0.0140 (0.0370)	0.0200 (0.0427)	0.0140 (0.0370)	0.0182 (0.0429)	−0.0166 (0.0351)
VideoGameEq	0.1021** (0.0416)	0.0308 (0.0372)	0.1022** (0.0416)	0.0307 (0.0375)	0.1005** (0.0417)	0.0348 (0.0355)
Instruments	0.0144 (0.0101)	0.0015 (0.0095)	0.0144 (0.0101)	0.0015 (0.0096)	0.0144 (0.0100)	−0.0016 (0.0092)
VideoCamEq	−0.0104 (0.0399)	0.0630* (0.0340)	−0.0104 (0.0399)	0.0631* (0.0340)	−0.0129 (0.0400)	0.0076 (0.0325)
TVEq	−0.0047 (0.0173)	0.0176 (0.0142)	−0.0047 (0.0174)	0.0176 (0.0142)	−0.0049 (0.0174)	0.0123 (0.0137)
PhotoEq	−0.0093 (0.0153)	0.0064 (0.0081)	−0.0093 (0.0153)	0.0064 (0.0081)	−0.0100 (0.0156)	0.0154 (0.0097)
PriceRestL	0.3848** (0.0458)		0.3848** (0.0458)		0.3859** (0.0460)	
QuantityRestL	0.4628** (0.1002)		0.4629** (0.1002)		0.4648** (0.1007)	
VarietyRestL	0.4707** (0.0465)		0.4708** (0.0466)		0.4729** (0.0468)	
DVDTVRestL	−0.1924 (0.1498)		−0.1925 (0.1501)		−0.1953 (0.1507)	

Table 5 continued

Variables	Model 1		Model 2		Model 3	
	Live	CD	Live	CD	Live	CD
SocialRestL	−0.1223 (0.1892)		−0.1226 (0.1909)		−0.1278 (0.1894)	
PriceRestCD		0.2720** (0.0333)		0.2720** (0.0333)		0.2867** (0.0310)
InternetHrs	0.0020 (0.0029)	−0.0131** (0.0030)	0.0020 (0.0029)	−0.0131** (0.0030)	0.0027 (0.0029)	−0.0083** (0.0028)
P2P	0.0384 (0.0525)	−0.0728 (0.0490)	0.0384 (0.0525)	−0.0729 (0.0492)	0.0424 (0.0528)	−0.1055** (0.0467)
ρ	0.1516**		0.1499**		0.0702**	
χ^2	2,604.1077		2,605.3686		3,088.4661	
p-value	0.0000		0.0000		0.0000	

Standard errors in parenthesis. $^{**}P < 0.05$, $^{*}P < 0.1$

4.1 The average music consumer

4.1.1 Socioeconomic variables

Only a limited set of the variables in this block are statistically significant. Gender is relevant when consuming popular music, either live or prerecorded. Results are quite robust and show that the probability of female participation decreases for all specifications, a finding that is coincident with that of Favaro and Frateschi (2007) for popular music univores. The magnitude of this *gender effect* is remarkably greater for prerecorded music, an activity that, in terms of information gathering, could be less risky than attending to a live concert. However, the social component of attendance, that is an individual's choice is driven by that of her partner, as Earl (2001) notes, could account for this observed gender effect. We will return to this later.

Age affects music consumption in a nonlinear fashion. Even though only its square is significant, a joint significance Wald test rejects the null hypothesis, defining an inverted U-shape for the overall effect of age on the consumption of prerecorded music while a distinctively negative effect for live attendance. Nevertheless, the point estimate for the net effect in both cases is negative.

The availability and the implicit cost of leisure time is relevant in all specifications when explaining attendance, while it does not affect purchases: being single has a positive effect on the probability of live music consumption, while raising children below 15 decreases it. No strong evidence was found for other variables related to time availability.

Labor market status affects both live and prerecorded music consumption mainly through an income effect. While working increases the likelihood of music consumption (except for live attendance in model 3), a feature that points out to a significant income effect, being a student has no significant effect on purchases or attendance. However, a one-tail test consistently supports a positive effect of being student on attendance, with a p-value of roughly 5.5%, and not on purchases.

Strikingly, being retired and househusband have both a positive effect on purchases. Even though surprising at first glance, intuition suggests that traditional consumers of prerecorded music, that is young people, are leaving the market as new consumption patterns emerge. This, in turn, alters the relative composition of demand with those consumers more averse to change or less technologically proficient being now overrepresented. In short, these consumers will show a tendency to stay in the market, a fact that is not fully captured by other technological variables such as those related to the internet use.

4.1.2 Cultural capital

Cultural capital is one of the most consistent determinants of cultural consumption, and by using it, we measure the degree of knowledge consumers acquire about a cultural manifestation, mainly through past consumption.

There are several aspects we bundle within this block. First, the highest education level that individuals have attained. Here, we find mixed results: attendance is

affected positively by individuals with vocational and university education, with high school level not affecting consumption. Nevertheless, and as it has been previously mentioned, it could be that education variables include an income effect.

Next, we consider active practice, courses, and participation in other cultural manifestations. The former affects consistently and in a positive way the probability of live and prerecorded music consumption. Having taken specific courses in arts does not affect the likelihood of consumption but marginally in live attendance for model 2. Participation in other cultural manifestations is definitely a trait of music consumers that show eclectic preferences: heritage, performing arts and cinema (both at theaters or at home) increase the probability of music consumption. Reading on the other hand is only linked to prerecorded music consumption.

TV hours and radio time may increase the amount of information consumers hold through the exposition of individuals to pop acts, releases, and performances. In this case, knowledge can be passively accumulated as individuals listen to the radio or watch television. Results point to radio exposure affecting positively attendance and purchases, while TV affects prerecorded music purchases.

One interesting finding is that the type of cultural activities individuals choose are reflected in the pattern of music consumption. Here, we resort to the distinction of relational and non-relational activities. Using Bruni and Stanca (2008) definition, relational goods are those in which the utility depends on interactions with others, and individuals enjoy it only if it is shared with others. From this standpoint, attending to live concerts can be considered a relational activity, while listening to prerecorded music a non-relational one. In our case, the probability of purchasing music is greater in consumers engaged in other non-relational activities: reading, TV watching, or watching movies all have a greater impact on purchasing. On the other hand, participation in the performing arts has a greater impact on live attendance.

4.1.3 Physical capital

To know or enjoy music, at least prerecorded music, one needs some basic equipment. However it has been noted that these variables also account, at least partially, for the household income. Therefore, we should not put too much emphasis on specific results.

Anyhow it is noteworthy that, apart from videogame consoles and the number of photo cameras, only music-related equipment was significant in all models, although not necessarily for both types of consumption.

4.1.4 Internet and file-sharing

With respect to the internet, our findings are robust for all the alternative specifications. Internet use (measured by weekly hours spent online) has a net negative effect on prerecorded music consumption, without affecting live attendance. This can be a consequence of the internet being a leisure alternative to listening to music. Nevertheless being user of file-sharing networks also affects quite consistently and in a negative fashion prerecorded music consumption. Overall

our findings seem to support a net negative effect of internet usage in the prerecorded music industry.

It is worth mentioning that this is not an individual but a joint test for crowding-out or exposition effects. What we observe is the net effect of both, and our findings are thus consistent with a positive external effect of the use of file-sharing on the consumption of prerecorded music. However, this seems to be offset by a negative, or substitution, effect.

More interesting is the fact that file-sharing has a positive but not significant effect on the Live variable. One could conclude that data do not support an effect of file-sharing on live attendance. Nevertheless, it is worth mentioning that a one-tailed test supports a marginal, i.e. 10% significance level, positive effect of P2P on attendance in model 3. This result suggests a marginal (positive) exposition effect of file-sharing on attendance. In other words, internet downloads can increase artists revenue through live performances.

4.1.5 Restrictions

Finally, we include individual restrictions to the frequency of participation. Those referring to the knowledge individuals have of the market—on prices, quantity or variety—are significant and positively related to consumption. Being better informed about market conditions makes consumers to reveal the limitations of their choice set due to facts such as price, limited supply or variety.

However, we did not find any significant effect of alternative consumption ways or social restrictions on live attendance. This in turn implies that we find no evidence of recorded media being substitute for live attendance, or of social gathering motivations to restrict average consumer decisions.

4.2 Frequent consumers

Table 5 shows the estimation results for frequent consumers, i.e. those that attended or purchased at least once in the last 3 months. In addition to the explanatory variables in Table 4, it includes a dummy to control for the term in which the survey was taken, as it can affect attendance and purchase behavior.

The first difference with results for the average consumer is a stronger gender effect: being female consistently decreases the likelihood of frequent attendance and consumption of popular music. It is also remarkable that the gender effect is now greater for attendance than for purchases.

Second, there are significant differences with regard to labor market classification. Workers have both a greater likelihood of frequent attendance and purchases. However, we find that being unemployed also affects both endogenous variables (but for model 3 that affects only attendance), while looking for the first job has a positive effect only on live attendance. We also find that being a student increases the likelihood of both activities except for model 3, in which the effect is significant for attendance only. Being a househusband increases the likelihood for both activities.

At first glance, these results might look as counterintuitive. However, two facts explain them. On the one hand, attendance may be affected by an individual's time constraint. Hence, individuals in any of the three mentioned categories (that is students, househusbands and unemployed either first time or not) will have plenty of time available which makes them more likely of being frequent attendees (and maybe music listeners). If this is the right explanation, overall the time effect offsets any negative income effect and pushes up frequent consumption of both live and prerecorded music.[14] The role of the time restriction is also apparent in the stronger effect of raising children.

On the other hand, these results show how the probability of attendance and purchase is affected relative to an individual exhibiting some specific trait. For live attendance, this means that those working or looking for a job, studying, or undertaking domestic tasks are more likely to attend to a live performance than those retired and individuals in the reference group, i.e. those in neither of the previous categories. A similar reasoning applies to recorded music.

Third, we find a significant impact of listening to the radio in purchases of prerecorded music, except for model 3, while TV hours now have a negative impact on attendance (maybe due to the non-relational activity impact of the former) and only in model 3 we see also a positive impact on purchases. If both TV and radio provide information about acts and concerts, the frequent consumer, unlike the average one, collects it mainly through the radio. This fits with TV catering to broader audiences, i.e. average consumers, while radio stations being more targeted.

Fourth, although the variable capturing alternative recorded consumption for live performances is non-significant (*DVDTVRestL*), a one-tailed test supports it being marginally negative. Therefore, we find a negative effect on attendance for frequent consumers who reveal that the availability of alternative recorded consumption substitutes actual live attendances. By and large, a segment of frequent consumers finds recorded media a suitable alternative for live attendances. For them, there is no other motivation besides the music itself, and the experience has to be considered as an information collecting activity. However, we must note that these are a small share of the sample (3.37% as per Table 3); hence, for most people, alternative motivations to attendance, which may include any of those in Earl's paper, are relevant.

Fifth, and finally, P2P effect is less clear for frequent consumers. However, a one-sided test supports a marginal negative effect, a result that suggests a weaker substitution effect for the frequent consumer. Besides, and contrary to the finding for the average consumer, there is no evidence of an exposition effect on attendance. Combining both pieces of information, we conclude that frequent consumers are less affected by file-sharing, what could point to specific traits in this segment such as a specific taste for music. In addition, one might expect frequent consumers to have a greater knowledge of music through repeated consumption, making exposition effects, i.e. discovery of music or acts, less likely.

[14] At this point, it is important to note that prerecorded music includes both physical media and digital downloads to computers or mobile phones.

4.3 The causal link and the "Bowie theory" revisited

We close the analysis of the results by comparing the three alternative specifications in Tables 4 and 5, and analyzing the observed degree of complementarity between both markets, for the average and the frequent consumer. The origin of these complementarities are to be found in the role of live performances as a way of promoting prerecorded music sales.

What Krueger (2005) labels as the *Bowie theory*, refers to the [weakening of the] direct causal link of the live market on the prerecorded one. Krueger's aim when proposing this theory was to explain the increase in live concert prices. If concerts are a source of demand for prerecorded music, it would be optimal for the latter to subsidize the former. As a consequence, actual concert prices would be less than those observed in the absence of complementarities.

As it has been mentioned, we link both markets indirectly, through the correlation coefficient ρ, and directly through a causal link. Two results emerge. First, note that even though profiles are different for live and prerecorded music consumers, estimates for ρ in Tables 4 and 5 show a significant correlation between the unobserved components in both equations. This means, among other things, that we cannot consistently estimate both equations independently, as there are unobserved correlated factors that drive demand in both sectors.

Second, we do not find evidence of a direct causal link from live attendance to prerecorded music demand. To test for it, we include live attendance in the prerecorded music equation, the specification labeled as model 2. Neither Table 4 nor 5 show significant results, an outcome that prevents us to reject the non-existence of a direct connection between attendance and purchases for both the average and the frequent consumer: empirical evidence does not support a higher probability of purchasing music as a consequence of attending to live concerts. This finding is consistent with Krueger's tentative explanation of the price dynamics in the live sector.

However, could causality be reversed? Model 3 checks for an inverse causal link from prerecorded music to live concert attendance. It is worth mentioning that in both tables we find positive and significant results for both average and frequent consumers, which means that prerecorded music consumers have a greater likelihood to attend live concerts. This effect is the strongest individual determinant for attendance in the case of the average consumer, while its effect is more modest for the frequent consumer.

Furthermore, once we account for the causality from CD to Live (model 3), the correlation coefficient between the unobserved heterogeneity is negative for average consumers and positive for frequent consumers. This positive correlation between frequent consumers of prerecorded and live music can be interpreted as a common unobserved trait present for both consumers that, using Prieto-Rodriguez and Fernandez-Blanco (2000) terminology, can be linked with a taste for music. A common trait that is not present for average consumers.

All these results suggest some interesting implications that are further explored next.

5 Conclusions and discussion

The main objective of this work has been to analyze the profile of the popular music consumer, considering both live and prerecorded popular music. We have carried out the empirical analysis by estimating a bivariate probit specification that, additionally, has allowed us to test hypotheses about the impact of the internet and file sharing on popular music consumption and the potential substitution and social motives for attendance. Furthermore, and in order to find some empirical evidence to Kruegers' *Bowie effect*, we have also accounted for the links between both markets due to correlated observable variables, correlated unobserved heterogeneity, and direct causal links.

Regarding our main objective, we have profiled the average and frequent music consumer. Empirical results show similarities between both types, being gender, age and the impact of active practice and cultural participation common covariates. However, there are significant differences between frequent and average consumers. First, the most striking difference is on the stronger role played by time constraints for live attendance for frequent consumers. This is interesting as cultural activities are usually regarded as time intensive. In this case, time availability becomes a major determinant of frequent consumption at a higher level than income.

Second, we found differences with respect to the potential substitution effect between live and prerecorded music for a share of the population. The demand for popular music of some frequent consumers is a demand for information, which can be catered by alternative means. For everyone else, the demand for live music has to be something else than the demand for information, as Earl (2001) suggested.

Third, we also find marginal evidence of a marginal P2P-exposition effect for average live consumers that is absent for frequent live consumers. This means that the exposition to prerecorded music through file-sharing networks leads to an increase of average, but not frequent, live attendance. Overall, we found some evidence of a negative impact of the use of the internet and file-sharing networks on the prerecorded music market, although it was weaker for the frequent consumer. This can be due to the internet competing with music as a new leisure activity but also may account for a net substitution effect which is not offset by potential positive external effects.

Regarding the existence of complementarities in both markets, the findings show evidence of a causal link between them, with prerecorded music consumption increasing the probability of live concerts attendance, while the latter does not affect music purchases in a significant way. This result was found for average and frequent consumers, although it was stronger in the former case. As we have already mentioned, this is consistent with and supports Kruegers' *Bowie theory*.

Apart from direct causality, we have looked at the correlation between consumers in both markets. Interestingly, when the full model is specified (model 3), frequent consumers show a common disposition to music, either live or prerecorded. This is consistent with frequent consumers being music lovers and being catered for by both markets. On the contrary, unobserved heterogeneity moves average consumers either to one market or the other. Let us consider the unobserved trait for the average consumer to be the social motivation for attendance, then individuals with

such a trait will be more likely of consuming live music and less likely of consuming recorded music. In other words, once we account for direct links between both markets and for all the covariates, the average consumer is either a consumer of live or prerecorded music.

Finally, Earl's main thesis, i.e. the difficulty to justify live music consumption as the demand for information, finds some support in our setup. The substitutability relationship between live and prerecorded music for frequent consumers, and the fact that average music consumers may be different depending on whether they consume prerecorded or live music, allows us to link motivations to unobserved features, a finding that could accommodate any of Earl's consumption motives. Nevertheless, we need to point out that the available data in our data set restricts us from a throughout test of all the implications in Earl (2001).

One potential drawback of our analysis is that it does not consider consumers of prerecorded music in a comprehensive perspective. In other words, the prerecorded music sector should include not only buyers but also users of prerecorded music no matter whether they buy it, copy it or download it for free from the internet. Perhaps attending to a concert does not increase the likelihood of buying prerecorded music, but it may increase the likelihood of listening to prerecorded music obtained by any means.

In order to check for the consistency of our findings, we run the same set of regressions changing the dependent variable prerecorded music purchase for prerecorded music user. This includes not only buyers but also those that do not buy because they copy music from friends or acquaintances, or download it. Estimation results do not vary in a significant way,[15] and, interestingly, the causality link is kept. There is no significant direct effect from live concert attendance to prerecorded music consumption, now either buying or getting it by any other means. Besides, evidence again suggests a strong causation from prerecorded music consumption to attendance.

This is an interesting result in itself, as it may give an economic rationale for recent changes in the sector. Whether in the past the prerecorded sector has subsidized the live music sector, results suggest that nowadays subsidization will be the other way round: from concerts to prerecorded music. With prerecorded music prices falling[16] and many artists giving away recorded music at concerts, the internet and other digital platforms, this seems to be the case.

This helps explain new contractual setups the recording industry has been implementing with new acts—and some established ones—in the the past few years. Examples are "360°" or all inclusive contracts. Its main feature is that these include rights not only for prerecorded music sales, but also for management, concert, merchandise and endorsement deals among other. These make sense from an anecdotal standpoint as record labels are trying to capture part of the revenue of most profitable segment of the music market. However, they also make sense from an economic standpoint by acknowledging that one sector is subsidizing the other.

[15] Results are not included for the sake of economy. However they are available on demand.

[16] Available data published by the *Spanish Phonographic Association* show that over the last decade sale cuts by value exceed those by amount.

Moreover, all-inclusive contracts help internalize the external effects that some activities in the music industry generate. When these are performed by different actors, a problem of misaligned incentives emerges, with the underprovision of those activities that generate positive external effects. Think of a record label promoting a new release; unless the label can capture part of the rents that it will create, and nowadays most of them will be related to touring, incentives will play against promotion. By coordinating different activities in the music market under one firm—be it a record label or a concert promoter—the external effect problem is ruled out and optimal outcomes can be achieved.

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References

Ateca-Amestoy, V. (2008). Determining heterogeneous behavior for theater attendance. *Journal of Cultural Economics, 32*(2), 127–151.

Black, G., Fox, M., & Kochanowski, P. (2007). Concert tour success in North America: An examination of the top 100 tours from 1997 to 2005. *Popular Music and Society, 30*(2), 149–172.

Bruni, L., & Stanca L. (2008). Watching alone: Relational goods, television and happiness. *Journal of Economic Behavior and Organization, 65*(3–4), 506–528.

DiMaggio, P. (1990). "A highbrow/lowbrow": The emergence of cultural hierarchy in America. *The Sociological Review, 38*(3), 608–612.

DiMaggio, P., & Ostrower, F. (1990). Participation in the arts by black and white americans. *Social Forces, 68*(3), 753–778.

Earl, P. E. (2001). Simon's travel theorem and the demand for live music. *Journal of Economic Psychology, 22*(3), 335–358.

Favaro, D., & Frateschi, C. (2007). A discrete choice model of consumption of cultural goods: The case of music. *Journal of Cultural Economics, 31*(3), 205–234.

Fisher, T. C. G., & Preece, S. B. (2003). Evolution, extinction, or status quo? Canadian performing arts audiences in the 1990s. *Poetics, 31*,69–86.

Greene, W. H. (2007). *Econometric analysis*. New Jersey: Prentice-Hall International.

Krueger, A. B. (2005). The economics of real superstars: The market for rock concerts in the material world. *Journal of Labor Economics, 23*(1), 1–30.

Levy-Garboua, L., & Montmarquette, C. (1996). A microeconomic study of theatre demand. *Journal of Cultural Economics, 20*(1), 25–50.

Maddala, G. S. (1983). *Limited-dependent and qualitative variables in econometrics*. Cambridge: Cambridge University Press.

Minor, M., Wagner, T., Brewerton, F., & Hausman, A. (2004). Rock on! an elementary model of customer satisfaction with musical performances. *Journal of Services Marketing, 18*(1), 7–18.

Montoro-Pons, J. D., & Cuadrado, M. (2008). Legal origin and intellectual property rights: An empirical study in the prerecorded music sector. *European Journal of Law and Economics, 26*(2), 153–173.

Oberholzer, F., & Strumpf, K. (2007). The effect of file sharing on record sales: An empirical analysis. *Journal of Political Economy, 115*(1), 1–42.

O'Hagan, J. W. (1996). Access to and participation in the arts: The case of those with low incomes/educational attainment. *Journal of Cultural Economics, 20*(4), 269–282.

Pareles, J. (2002). *David Bowie, 21st-century entrepreneur*. New York Times.

Peterson, R. A. (1997). The rise and fall of highbrow snobbery as a status marker. *Poetics, 25*(2–3), 75–92.

Peterson, R. A. (2005). Problems in comparative research: The example of omnivorousness. *Poetics, 33*, 257–282.

Peterson, R. A., & Kern R. M. (1996). Changing highbrow taste: From snob to omnivore. *American Sociological Review, 61*(5), 900–907.

Prieto-Rodriguez, J., & Fernandez-Blanco, V. (2000). Are popular and classical music listeners the same people? *Journal of Cultural Economics, 24*(2), 147–164.

Seaman, B. A. (2006). *Empirical studies for demand in the performing arts*. North Holland: Handbook of the Economics of Art and Culture.

The Economist. (2007, July 5). *A change of tune*. Print edition.

The Economist. (2010, January 8). *A union of pariahs. The ticketmaster-live nation merger*. Print edition.

Towse, R. (1992). *The earnings of singers: An economic analysis. Cultural Economics*. Springer: Heidelberg

Throsby, D. (1994). The production and consumption of the arts: A view of cultural economics. *Journal of Economic Literature, 32*(1), 1–29.

van Eijck, K. (2001). Social differentiation in musical taste patterns. *Social Forces, 79*(3), 1163–1185.

Zentner, A. (2006). Measuring the effect of file sharing on music purchases. *Journal of Law and Economics, 49*(1), 63–90.